



## Article

# Durability of Aircraft Gas Turbine Mainshaft Ball Bearing Applying Surface and Subsurface Survival

Guillermo Enrique Morales-Espejel <sup>1, 2)\*</sup>, Matthew Misiaszek<sup>3)</sup>, Thomas Lengagne<sup>3)</sup> and Liang Guo<sup>1)</sup>

<sup>1)</sup> SKF Research and Technology Development, Meidoornkade 14, 3992 AE Houten, The Netherlands

<sup>2)</sup> Université de Lyon, INSA-Lyon CNRS LaMCoS UMR5259, France

<sup>3)</sup> SKF Aerospace, Product Development & Testing, Chateaufort Sur Isère, France

\*Corresponding author: Guillermo Enrique Morales-Espejel (Guillermo.Morales@skf.com)

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## Abstract

A previously published model that separates surface from subsurface survival is adapted for high temperature bearing steels and it is applied in life calculations for aircraft gas turbine mainshaft rolling bearings. This approach shows more realistic calculated life than current practices with the use of the  $a_{23}(\frac{C}{p})^p$  method. Besides this, the proposed approach incorporates the effects of high temperature in the reduction of the fatigue limit, which is reflected in the life calculations, this can discriminate among steels working at high temperatures. As shown previously, with the separation of surface from subsurface, the effects from fatigue are quantified in the two separated zones and they can be compared, showing the zone of higher fatigue risk. Therefore, the proposed approach shows advantages in comparison with the current life calculation method used in the aircraft industry.

## Keywords

bearing life, temperature effect, aeroengine bearings, fatigue limit

## 1 Introduction

The durability of aircraft gas turbine mainshaft bearings has been studied in the past by Ioannides et al. [1, 2] by applying a new life model at the time. This model introduces the fatigue limit of the material and proper lubrication and contamination models. It is based directly on stresses rather than loads (stress-based model). The results obtained by this model were compared with the traditional calculation method used by the aircraft industry, based on the material ( $a_2$ ) and lubrication factor ( $a_3$ ), model incorporated in the computer program SHABERTH. The results show that the new model in the references would predict much higher bearing lives than the SHABERTH model. It must be noted that this new model [1, 2] derived in the current life calculation method was adopted in the ISO 281 in the year 2007, when reduced to a load-based model. At that time these results raised questions in the rolling bearing and aircraft communities, as can be seen in the discussions shown in [2], mainly with potential safety concerns among the detractors of the fatigue limit concept. More recently actual fatigue data which prove the existence of this property beyond any doubt and with an updated definition of fatigue limit in bearing steels were presented [3]. Gloeckner and Sebald [4] introduce a new

calculation method based on the ISO 281 but addressing some aspects of what this model still disregards, like steels used in aero and effect of high temperatures. They do this with the use of multiplication factors which they apply to the ISO 281 method. Yu et al. [5] focus on the way damage is accumulated in a duty cycle. They introduce a non-linear method in contrast with the previous linear methods used in [1, 2] like the cubic average law or the Palmgren-Miner rule.

After the update of the ISO 281 in 2007 not much has happened in the development of “industrial” bearing life models, although research has been very active. Sadeghi et al [6] give a complete overview of the different bearing life models from the more industry-applied ones to the academic ones. Subsequent to this review, among the few bearing life models recently introduced for industry usage is the model that separates surface from subsurface survival [7] and its later applications [8-11]. This model has the ability to handle different failure modes originated at the surface of the rolling bearings or beneath their surface. Therefore, it is more flexible as a result of being linked to the physics of the damage. In mainshaft aircraft gas turbine bearings most of the damage is originated at the surface, as in general these bearings are considered less heavily loaded when compared to other applications. They are,

therefore, a natural application for the model in [7]. However, as discussed in [4] one important feature missing in most of the current bearing life models is the effect of high temperatures typical in this application. This effect has influence in the bulk of the bearing as much as its surface, different from the case in [11] where the effect of high surface temperature was considered only in a local effect produced by frictional heating. In the mainshaft aircraft gas turbine application the whole bearing operates at temperatures close to 200°C.

Therefore, in this paper the surface and subsurface survival bearing life model introduced in [7] is adapted to consider the effects of bulk high temperature in two aero bearing steels with a modification of the fatigue limit. The model is then compared to bearing lives measured in high-temperature, high-speed endurance tests. The application of the rolling contact fatigue (RCF) model is also limited by the change of failure mode towards seizure. This limit is obtained from an anti-seizure minimum life test, conducted in extreme conditions of load and speed.

## 2 Bearing life model

### 2.1 Surface and subsurface survival

Only a brief introduction of the original model described in [7] will be given here for the sake of completeness. However, the reader is encouraged to go to the reference for the details. The derivation of the model follows simply the Weibull [12] rule of reliability, which basically says that the reliability (or probability of survival) of a structure is the multiplication of the reliability of each of its components, thus:

$$S = S_1 S_2 \cdots S_n \quad (1)$$

Which becomes:

$$\ln(S) = \ln(S_1) + \ln(S_2) \cdots + \cdots \ln(S_n) \quad (2)$$

Considering only two important parts in the structure, one being the surface and the other being the subsurface, Eq. (2) is reduced to:

$$\ln(S) = \ln(S_{sur}) + \ln(S_{sub}) \quad (3)$$

The reliability then can be related to stress integrals, material parameters and life by introducing the material degradation functions, following [7], thus:

$$L_{10} = \frac{\left[\ln\left(\frac{1}{0.9}\right)\right]^{1/e}}{u} \left[ \bar{A} \int_{V_v} \frac{(\sigma_v - \sigma_{u,v})^c}{z^h} dV_v + B \int_{V_v} \frac{(\sigma_v - \sigma_{u,v})^c}{z^h} dV_v \right]^{-1/e} \quad (4)$$

Where one integral corresponds to the subsurface and the other to the surface. For a shallow layer of thickness  $\hat{h}$  representing the surface, the second integral can be reduced to an area integral, with the exponent  $h = 0$ , like in [7].

$$L_{10} = \frac{\left[\ln\left(\frac{1}{0.9}\right)\right]^{1/e}}{u} \left[ \bar{A} \int_{V_v} \frac{(\sigma_v - \sigma_{u,v})^c}{z^h} dV_v + \bar{B} \int_A (\sigma_s - \sigma_{u,s})^c dA \right]^{-1/e} \quad (5)$$

With  $\bar{B} = \hat{h}B$

In Eq. (5) the first integral corresponds to the subsurface survival and the second integral to the surface survival. Notice that as explained in [7], this model assumes equal Weibull slopes in the surface and subsurface. The reason is a practical one from the solution point of view, to avoid an iteration process. The model at the end is calibrated with endurance tests

in its constants  $\bar{A}$  and  $\bar{B}$ .

In reference [7] the ratio of surface and subsurface integrals called “relative surface fatigue index” is introduced as:

$$RSF = \frac{I_s}{I_s + I_{ss}} \quad (6)$$

Where:

$$I_{ss} = \bar{A} \int_{V_v} \frac{(\sigma_v - \sigma_{u,v})^c}{z^h} dV_v \quad (6a)$$

And

$$I_s = \bar{B} \int_A (\sigma_s - \sigma_{u,s})^c dA \quad (6b)$$

Notice that when  $RSF \rightarrow 1$  then the surface integral dominates the life of the bearing population.

### 2.2 Bulk temperature effect

When hardened bearing steel is heated even below its stabilization temperature, it starts to lose fatigue strength and this is reflected in a reduction of the steel fatigue limit, as shown in [13]. In the reference two hardened steel Aero grade 52100 VAR (AMS 6444) and AMS M50 VIM/VAR (AMS 6491) and standard hardened bearing steel ASTM 52100 are studied using ultrasound push-pull in very-high-cycle-fatigue (VHCF) conditions exceeding  $10^{10}$  number of load cycles with three different temperatures, 22°C, 200°C and 300°C. Other researchers have tried to link the fatigue limit to the hardness [14, 15], naturally at higher temperatures hardness will be reduced and this can be connected indirectly to a temperature-induced reduction of fatigue limit. Finally, modelling has also been carried out to study the importance of different factors in the fatigue limit of aero bearing steels [16].

In this paper, the work of [13] in the measurement reduction of fatigue limit for bearing steels will be followed. The measured results for each steel will be curved-fitted in simple equations and this equation will be integrated in the model (5) for both surface and subsurface. In reference [13] a detailed analysis of the failures obtained is carried out, here only subsurface failures are considered as corresponding to the fatigue limit defined in ISO 281. Another important aspect from this work is that the fatigue limit is measured in a push-pull test. However, in rolling bearing the fatigue limit is expressed as an orthogonal shear stress. Given the consideration that the fatigue limit should not be influenced by crack propagation aspects (only initiation) it is possible to convert a push-pull test result into a shear stress fatigue limit as indicated in [3] and explained below.

Table 1 shows a summary of the read values from the plots of Fig. 5 of reference [13], representing the fatigue limit in push-pull conditions of the different steels at different temperatures. The values have been obtained considering  $5 \times 10^{10}$  number of cycles in all cases. In the cases where there are no failures at that number of cycles extrapolation of the mean line was applied. With this data, curve-fitted equations were obtained:

For hardened AMS 52100 VAR:

$$\sigma_u = 0.0041T^2 - 3.1664T + 820.7707 \quad (7)$$

For hardened AMS M50 VIM/VAR:

$$\sigma_u = 0.0011T^2 - 0.7699T + 608.3857 \quad (8)$$

In Eqs. (7, 8) the temperature is expressed in Celsius degrees within the temperature range 22°C to 300°C and the fatigue limit given in MPa. As explained in [3] to correlate this

Table 1 Compilation from [13] of average measured fatigue limit at different temperatures

| Material        | $\sigma_u$ @22°C,<br>[MPa] | $\sigma_u$ @200°C,<br>[MPa] | $\sigma_u$ @300°C,<br>[MPa] |
|-----------------|----------------------------|-----------------------------|-----------------------------|
| AMS 52100 VAR   | 753                        | 352                         | 241                         |
| ASTM 52100      | 700                        | 347                         | 212                         |
| AMS M50 VIM/VAR | 592                        | 500                         | 480                         |

push-pull stress to equivalent shear stress it needs to be divided by  $\sqrt{3}$ . Also [3] shows push-pull measurements under ambient temperature for hardened ASTM 52100 which fluctuate around 700 MPa.

The fatigue criterion used in this paper is the orthogonal shear stress amplitude, thus in Eq. (5) the stresses  $\sigma$  represent orthogonal shear stress amplitude. In relation to the fatigue limits in Eq. (5), the same value is considered for the surface and subsurface. Thus, introducing proper nomenclature, Eq. (5) can then be rewritten as:

$$L_{10} = \frac{\left[ \ln \left( \frac{1}{0.9} \right) \right]^{1/e}}{u} \left[ \bar{A} \int_{V_v} \frac{(\tau_v - \tau_u)^c}{z^h} dV_v + \bar{B} \int_A (\tau_s - \tau_u)^c dA \right]^{-1/e} \quad (9)$$

With  $\tau_u = \sigma_u/\sqrt{3}$  and given by Eqs. (7, 8) depending on the material.

This equation can be solved in the way described in [7], by either using a partial lubrication model for the surface integral or by using a simplified curve-fit. The purpose of the present paper is not to detail the solution method, since it has been discussed elsewhere, but rather to show and discuss the results related to the bulk temperatures.

### 3 Bearing endurance testing and comparison with model

Full endurance tests were conducted for each material variant at so-called “low” and “high” temperatures, representing 90°C and 200°C outer ring temperature, respectively. The endurance tests were carried out in a modified SKF R2 rig with schematics represented in Fig. 1. The modifications relate to the addition of electrical heating and temperature control for the high-temperature test, to maintain the outer ring to a constant temperature of 200°C throughout the test duration.

Table 2 shows the operating conditions applied in the “low” and “high” temperature endurance tests per material carried out for model validation. All endurance tests were conducted with only axially loaded angular contact ball bearings ISO designation 7209, in all cases the rings were made of the corresponding material and heat treatment and the balls in all cases were made of AMS M50 VIM/VAR material and brass cages were used. All tests correspond to clean, full-film conditions as the goal of the test is to measure the effect of bulk material temperature on fatigue limit. Figure 2 shows the corresponding contact pressures for the heaviest loaded ball in the inner ring (IR) and the outer ring (OR).

In both test conditions most of the failures appeared in the inner rings, however some few failures appeared in the outer rings. Only 2 balls failed with spalls in the high-temperature test. Figure 3 shows a Weibull plot with all test results of the different steel variants at the two testing temperatures. As a reference or baseline results with standard ASTM 52100 material at “low” temperature are also included in the plot.

Table 3 summarizes measured and calculated lives of the different tested variants. The life results are normalized with respect to the shortest individual bearing life in all tests.

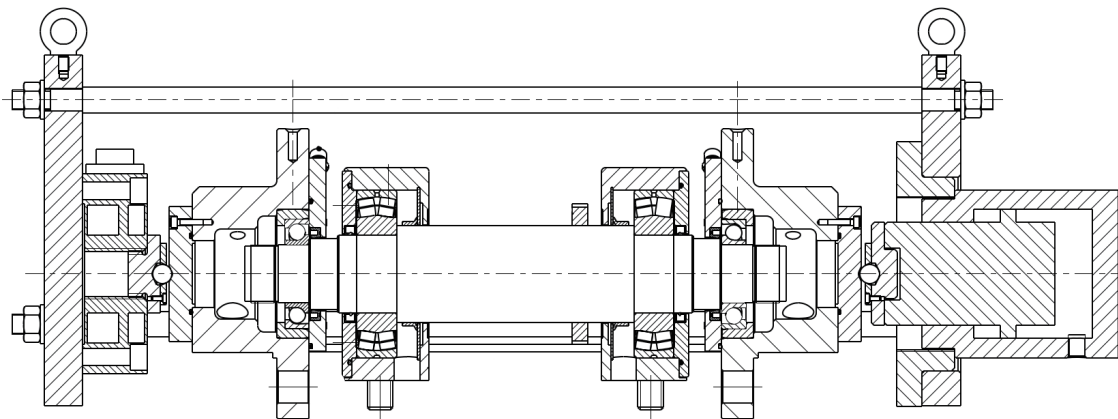


Fig. 1 Schematics of the SKF R2 endurance testing rig, where radial and axial load can be applied to the “smaller” test bearings

Table 2 Operating conditions for the endurance tests at “low” and “high” temperatures

| Bearing | Outer ring temperature, [°C] | Axial load, [kN] | Radial load, [kN] | Rotational speed, [rpm] | Kappa, [-] | Oil                 |
|---------|------------------------------|------------------|-------------------|-------------------------|------------|---------------------|
| 7209    | 90                           | 50               | 0                 | 8000                    | 2          | ISO VG 68 (mineral) |
| 7209    | 200                          | 50               | 0                 | 8000                    | 2          | ISO VG 1000 (ester) |

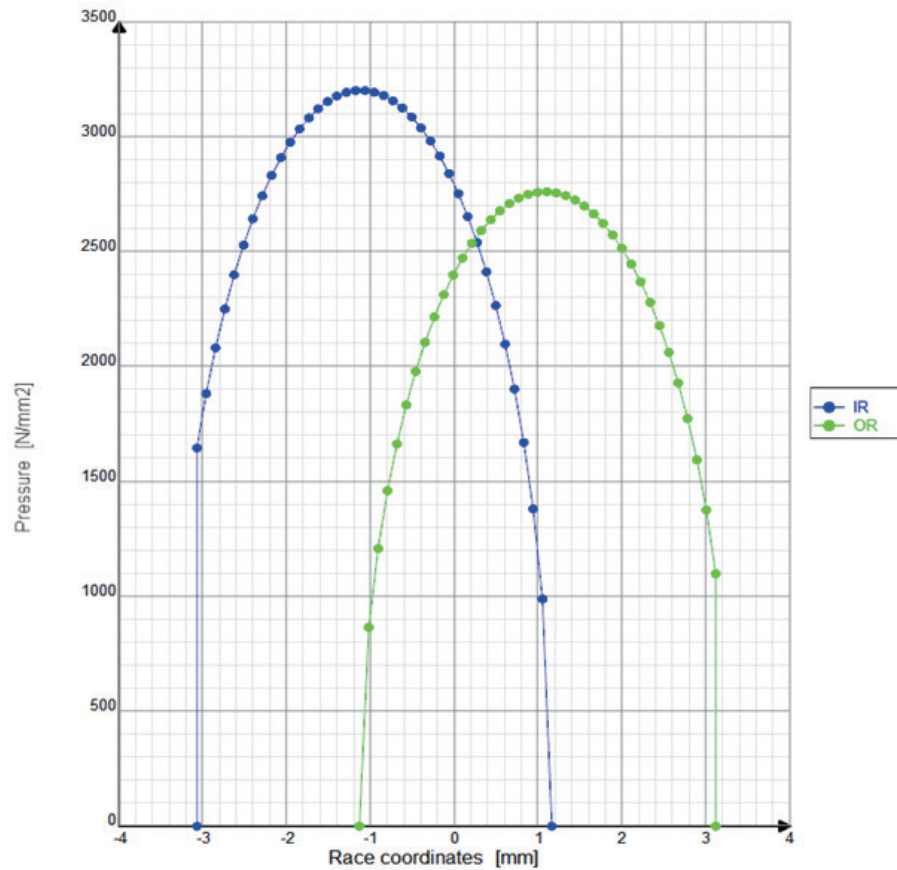


Fig. 2 Contact pressures for the heaviest loaded ball in the inner ring (IR) and the outer ring (OR). In this case (pure axial load) all balls bear the same load

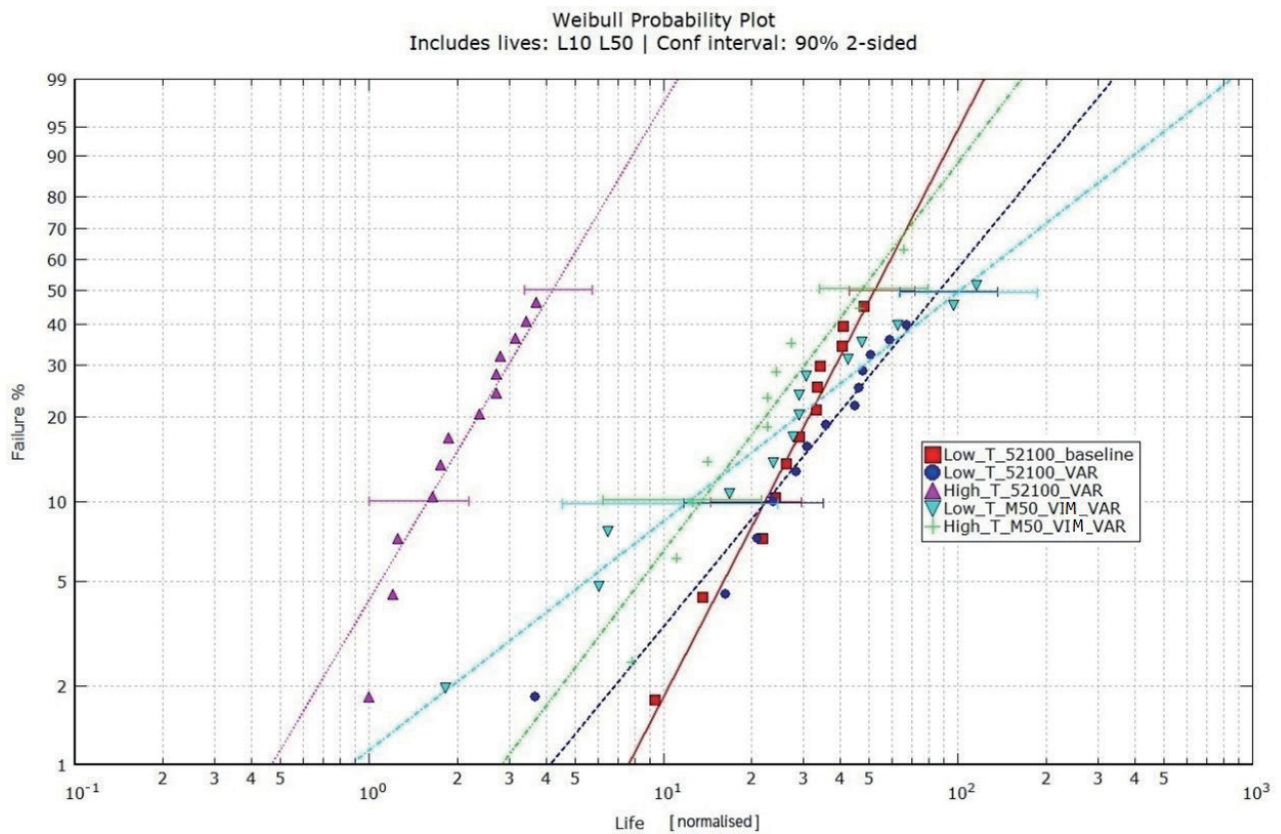


Fig. 3 Weibull plot of the test results at “low” and “high” temperature of all tested bearings



Table 3 Summary of measured and calculated lives of the different tested variants

| Variant         | Measured life<br>$L_{10,50r}$ [normalized] | Calculated life<br>$L_{10r}$ [normalized] | Outer ring<br>temperature, [°C] |
|-----------------|--------------------------------------------|-------------------------------------------|---------------------------------|
| AMS 52100 VAR   | 22.40                                      | 15.2                                      | 90, "low"                       |
| AMS 52100 VAR   | 1.58                                       | 1.17                                      | 200, "high"                     |
| AMS M50 VIM/VAR | 12.12                                      | 15.02                                     | 90, "low"                       |
| AMS M50 VIM/VAR | 13.45                                      | 7.6                                       | 200, "high"                     |

For the aero materials the effects of temperature in the fatigue limit have been applied in the calculations and in all cases the calculated values are nearly equal or below the  $L_{10,50}$  measured life, which is normal practice in bearing life calculations for safety. Only for the case of AMS M50 VIM/VAR at low temperature is the model slightly higher but still within a reasonable range. Finally, Fig. 4 shows the Weibull plot results compared with the model results (vertical lines).

#### 4 Anti-seizure test for model limits

This test has the objective to verify that in general the bearings will avoid seizure failure mode [17, 18] given the

severity of operating conditions (combination of high speed and high load) that this application requires. If this is verified, this will ensure that the fatigue damage model will prevail and therefore that a fatigue-based model can be applied.

The test campaign is based on a no-failure minimum life endurance strategy based on statistical principles. The objective is that all bearings reach several times the calculated  $L_{10}$  life with the test operating conditions. This will ensure a minimum life  $L_{10}$  for the bearing population within a certain reliability. Populations of bearings in both executions steel-steel and ceramic (hybrid) bearings were tested. The targeted operating conditions are summarized in Table 4 for both tests, steel-steel and ceramic (hybrid) bearings. The average measured

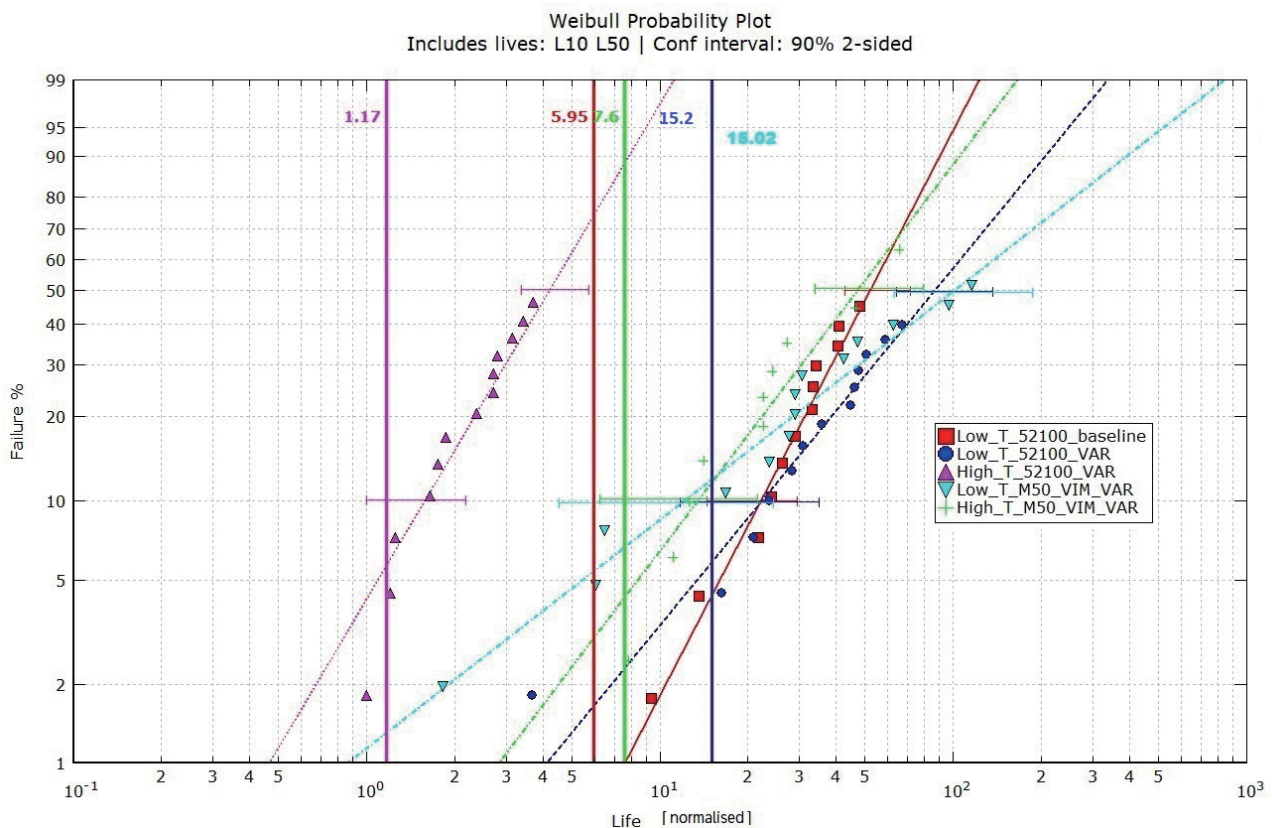


Fig. 4 Normalised Weibull plot of the test results at "low" and "high" temperature of all tested bearings compared with the model results (vertical lines) of Table 3

Table 4 Anti-seizure test conditions for the steel-steel and hybrid bearings, in both cases assuming a Weibull slope of 1 and 99% confidence

| Bearing execution | Contact pressure [GPa] | Rotational speed [ $nd_m$ ] | No. bearings tested | Suspension time [hrs] |
|-------------------|------------------------|-----------------------------|---------------------|-----------------------|
| Steel-steel       | 3                      | $2.3 \times 10^6$           | 10                  | 50                    |
| hybrid            | 3.5                    | $2.9 \times 10^6$           | 6                   | 40                    |

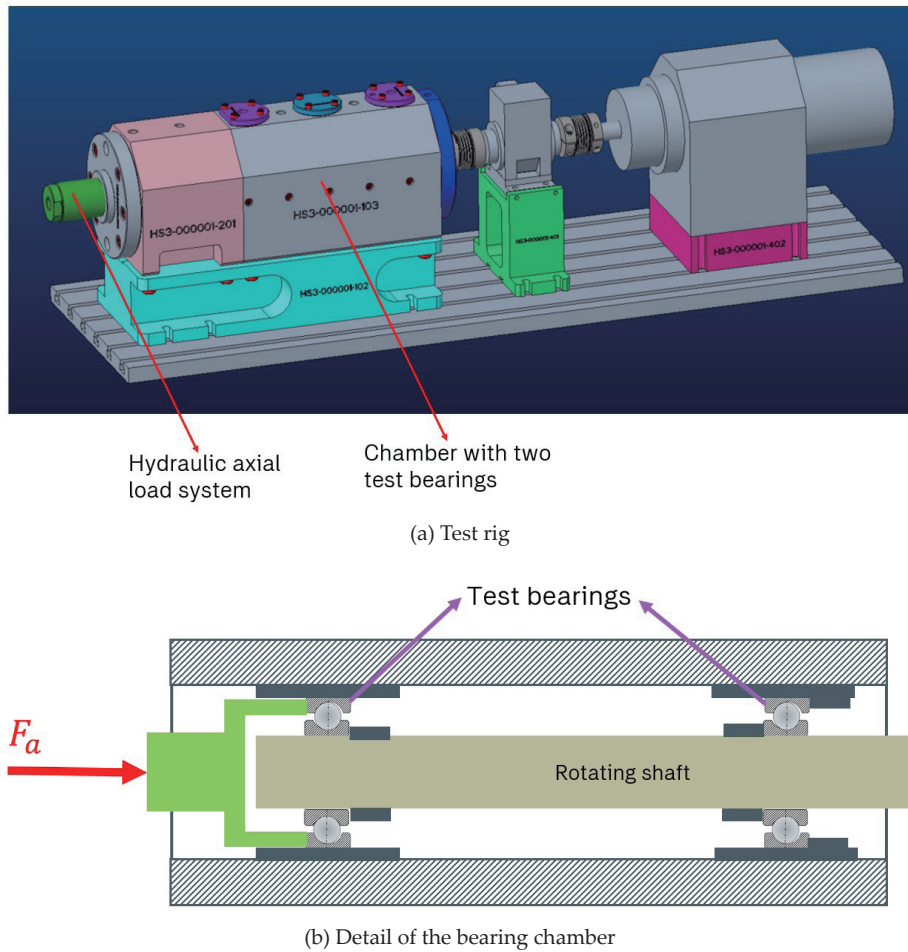


Fig. 5 Schematics of the Anti-seizure test rig and the two axially loaded test bearings

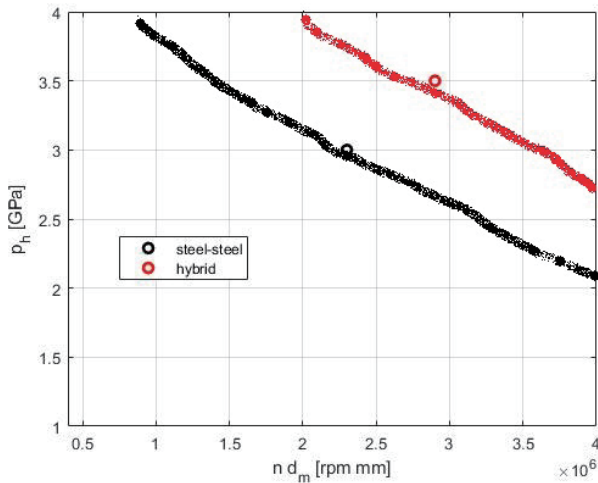


Fig. 6 Plotted operating conditions of the anti-seizure tests (circles) with approximately drawn line limits of the transition diagrams to limit the fatigue and the seizure failure mode similar to [17]. For operating conditions with higher pressure or speed (exceeding the above lines) seizure might occur

temperature in the outer ring during the test for the hybrid bearings was around 150°C and for the steel-steel bearings of 154°C.

The selected bearing is an aerospace variant of a 4-contact point ball bearing with under-race lubrication in the inner ring, similar external basic dimensions of the deep groove ball bearing ISO 6009. The bearings are lubricated with jet oil VG 32. A schematic of the test rig used for this test is shown in Fig. 5. Two axially loaded bearings are tested at a time.

The design of the anti-seizure test is based on the minimum-life concept which assumes a known Weibull slope value. In this case a slope of 1.0 was chosen. As an example, for the hybrid bearings the calculated  $L_{10}$  is about 5 hrs. If the test is successful (running all bearings until the suspension time without failures) it means that with 99% confidence the life of the population is at least  $8 \times L_{10}$ . In the present case it resulted in testing 6 bearings for 40 hours as suspension time. Both tests were successful, and the operating conditions of Table 4 are validated for minimum life. The results can be summarised in Fig. 6, referring to the contact pressure vs. rotational speed transition diagrams, as introduced in [17]. In this figure, approximated transition diagrams are drawn to limit the fatigue from the seizure failure modes.

## 5 Aircraft engine application example

Ioannides et al. [1] present failure incident statistics in a

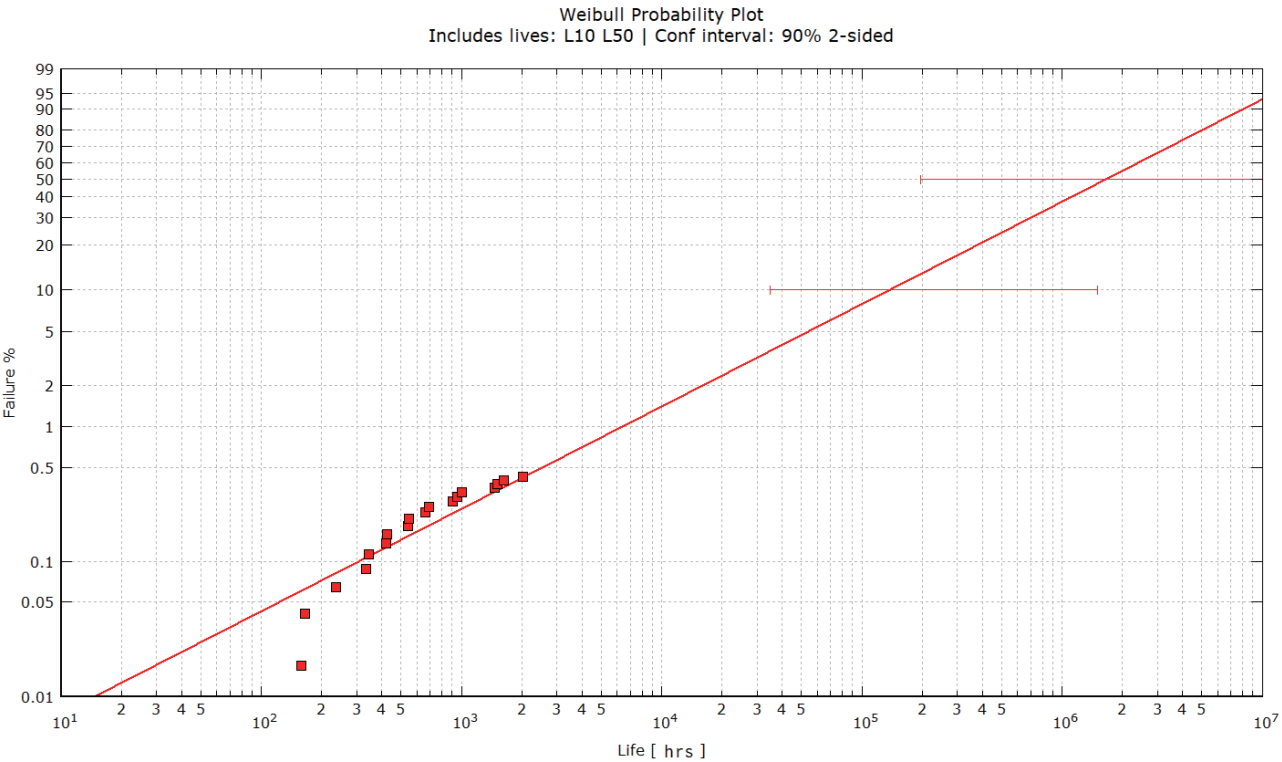


Fig. 7 Weibull plot corresponding to the failure incidents presented in [1] with a sample of 4173 bearings

period exceeding 10 years, showing 18 failures of about 4173 running bearings. The Weibull plot corresponding to this sample is shown in Fig. 7. Main bearing geometry aspects for this case are summarized in Table 5.

The duty cycle applied to these bearings according to [1] is described in Table 6. Life calculations with the SHABERTH computer program (normally used mainly in the USA aero applications) are given in Table 7 as taken from reference [1]. This gives for ASTM 52100 hardened steel a total calculated life of 230 hrs. However, for the considered material in the reference AMS M50 VIM/VAR, a factor  $a_2 = 6$  is applied given

Table 5 Mainshaft ball bearing internal dimensions of the case described in [1]

| Design Parameter                    | Dimension  |
|-------------------------------------|------------|
| Nominal pitch diameter              | 190 mm     |
| Ball diameter                       | 23.02 mm   |
| Number of balls                     | 20         |
| Raceway groove radius/Ball diameter |            |
| Inner raceway                       | 0.52       |
| Outer raceway                       | 0.52       |
| Nominal contact angle               | 32 degrees |
| Raceways temperatures               |            |
| Inner raceway                       | 121°C      |
| Outer raceway                       | 149°C      |
| Surface roughness, Ra               |            |
| Inner raceway                       | 0.102 μm   |
| Outer raceway                       | 0.102 μm   |
| Ball                                | 0.013 μm   |

the calculated life with SHABERTH of ~ 1435 hrs.

Now, to simulate this case with the present approach which does not use the lubrication factor  $a_3$ , some considerations need to be discussed. Heemskerk [19] formalized the use of the viscosity ratio  $\kappa$  rather than the film thickness parameter  $\Lambda$  to link lubrication quality effects on bearing life. This idea was then adopted by ISO 281: 2007 and since then modern life calculation methods consider the viscosity ratio instead of the film thickness parameter. In [20] one of the current

Table 6 Mainshaft ball bearing duty cycle, according to [1]

| Condition No. | Time, [%] | Speed, [rpm] | Axial load, [N] | Radial load, [N] |
|---------------|-----------|--------------|-----------------|------------------|
| 1             | 36.7      | 9319         | 1780            | 5607             |
| 2             | 0.7       | 9691         | 2581            | 4717             |
| 3             | 2.7       | 12015        | 31577           | 3605             |
| 4             | 1.5       | 10464        | 12469           | 4050             |
| 5             | 7.6       | 11464        | 25174           | 4828             |
| 6             | 5.2       | 11722        | 30491           | 4495             |
| 7             | 2.3       | 11909        | 35097           | 3827             |
| 8             | 7.5       | 11204        | 19860           | 4828             |
| 9             | 5.8       | 11039        | 17213           | 4027             |
| 10            | 2.1       | 11485        | 22143           | 4828             |
| 11            | 16.9      | 11090        | 15682           | 4606             |
| 12            | 1.3       | 12990        | 48109           | 2626             |
| 13            | 5.0       | 13100        | 63502           | 2715             |
| 14            | 2.9       | 13011        | 48011           | 2826             |
| 15            | 1.3       | 12993        | 47642           | 2826             |
| 16            | 0.5       | 12965        | 41639           | 2826             |

Table 7 SHABERTH  $a_2$   $a_{23}(\frac{\sigma}{\sigma_p})^p$  results in bearing life per condition of the duty cycle of Table 6 as given in [1] for ASTM 52100 hardened steel

| Condition No.               | Life, inner ring, [hrs] | Life, outer ring, [hrs] | Bearing life, [hrs] | Lubrication factor $a_3$ , Outer ring | Lubrication factor $a_3$ , Inner ring |
|-----------------------------|-------------------------|-------------------------|---------------------|---------------------------------------|---------------------------------------|
| 1                           | 6267                    | 15340                   | 4721                | 1.13                                  | 1.81                                  |
| 2                           | 6849                    | 22510                   | 5537                | 1.22                                  | 1.86                                  |
| 3                           | 339                     | 444                     | 206                 | 1.48                                  | 1.96                                  |
| 4                           | 2199                    | 5487                    | 1665                | 1.36                                  | 1.92                                  |
| 5                           | 528                     | 760                     | 333                 | 1.42                                  | 1.93                                  |
| 6                           | 367                     | 473                     | 221                 | 1.44                                  | 1.94                                  |
| 7                           | 268                     | 313                     | 156                 | 1.45                                  | 1.93                                  |
| 8                           | 850                     | 1464                    | 574                 | 1.41                                  | 1.94                                  |
| 9                           | 1156                    | 2281                    | 818                 | 1.44                                  | 1.95                                  |
| 10                          | 686                     | 1136                    | 457                 | 1.44                                  | 1.94                                  |
| 11                          | 1327                    | 2906                    | 969                 | 1.42                                  | 1.94                                  |
| 12                          | 123                     | 128                     | 67                  | 1.54                                  | 1.98                                  |
| 13                          | 63                      | 55                      | 31                  | 1.50                                  | 1.96                                  |
| 14                          | 123                     | 128                     | 67                  | 1.54                                  | 1.98                                  |
| 15                          | 122                     | 126                     | 66                  | 1.54                                  | 1.98                                  |
| 16                          | 171                     | 199                     | 98                  | 1.56                                  | 1.99                                  |
| Palmgren-Miner Overall Life |                         |                         | 230.1               | (considering revolution fractions)    |                                       |

authors discusses the advantages of this change, which can be summarized in the consideration of elastic roughness deformation, inlet heat sharing and kinematic starvation effect (high speeds) plus built-in safety factors. Therefore, from the data given in Ioannides et al. [1] the corresponding  $\kappa$  value can be determined in the following way.

In the method used by SHABERTH, the factor  $a_3$  is linked to  $\Lambda$  by the plot given in Bamberger et al. [21], re-plotted from digitized data in Fig. 8. In order to link the  $a_3$  factor to  $\kappa$ , the current approximation from ISO 281:2007 can be used,  $\kappa \approx \Lambda^{1.3}$ .

First, an equivalent  $a_3$  factor is calculated for each ring with the data of Table 7, and then it is averaged to get the “bearing”  $a_3$  this gives 1.6. With this value and Fig. 8 a value  $\Lambda = 1.784$  is

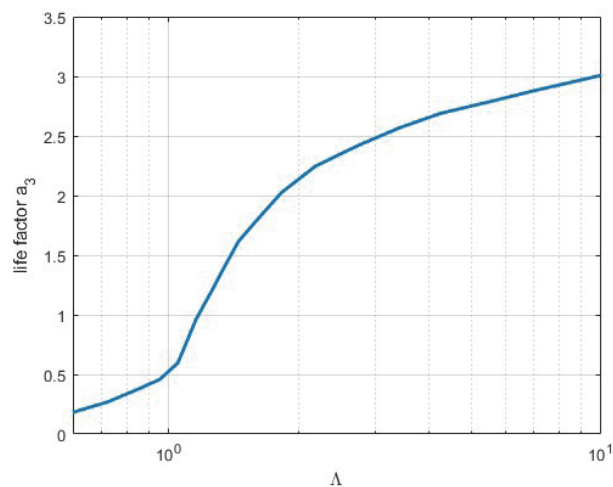


Fig. 8 Life factor  $a_3$  plot digitized from the original book of Bamberger et al. [21] used by the program SHABERTH

obtained. Therefore,  $\kappa \approx 2.14$  is calculated.

Following the same approach as in reference [1] in the application of “improved life theory” an equivalent operating condition is set for the load cycle of Table 6 is given in the reference: Thrust load 31132 N, 4806 N radial load and 10755 rpm inner ring speed. Under these conditions and  $\kappa = 2.14$  a calculated life for the bearing population made of AMS M50 VIM/VAR hardened steel with the here presented model results in  $L_{10} = 29321$  hrs. Figure 9 shows the comparison of this calculation with the experimental results of Fig. 7, for reference the SHABERTH result, as given in [1] is also included.

This result shows that at least the model is consistent with the experimental results and as discussed in [22] life bearing model results should always be placed to the left-hand side of the  $L_{10.50}$  percentile of the measure life to be conservative. However, since in this example the working temperatures of the bearings are not extreme high, see Table 5 (maximum 149°C), then this temperature will not produce significant differentiation in performance when varying the fatigue limit with temperature, the differentiation will be noticed at higher temperatures (maintaining the same  $\kappa$  value to remove the effect on lubrication).

This is shown in Fig. 10, where the life (or life ratio) has been plotted when varying bearing temperature. There are two cases in one the steel fatigue limit is affected by the temperature and in the other (reference) this effect is not considered.

## 6 Discussion

In the present article, the bearing life model developed earlier [7] that separates surface and subsurface survival has been adapted now to account for high temperature bearing steels such as the ones used in mainshaft aero-engine applications by considering the effects of temperature in the



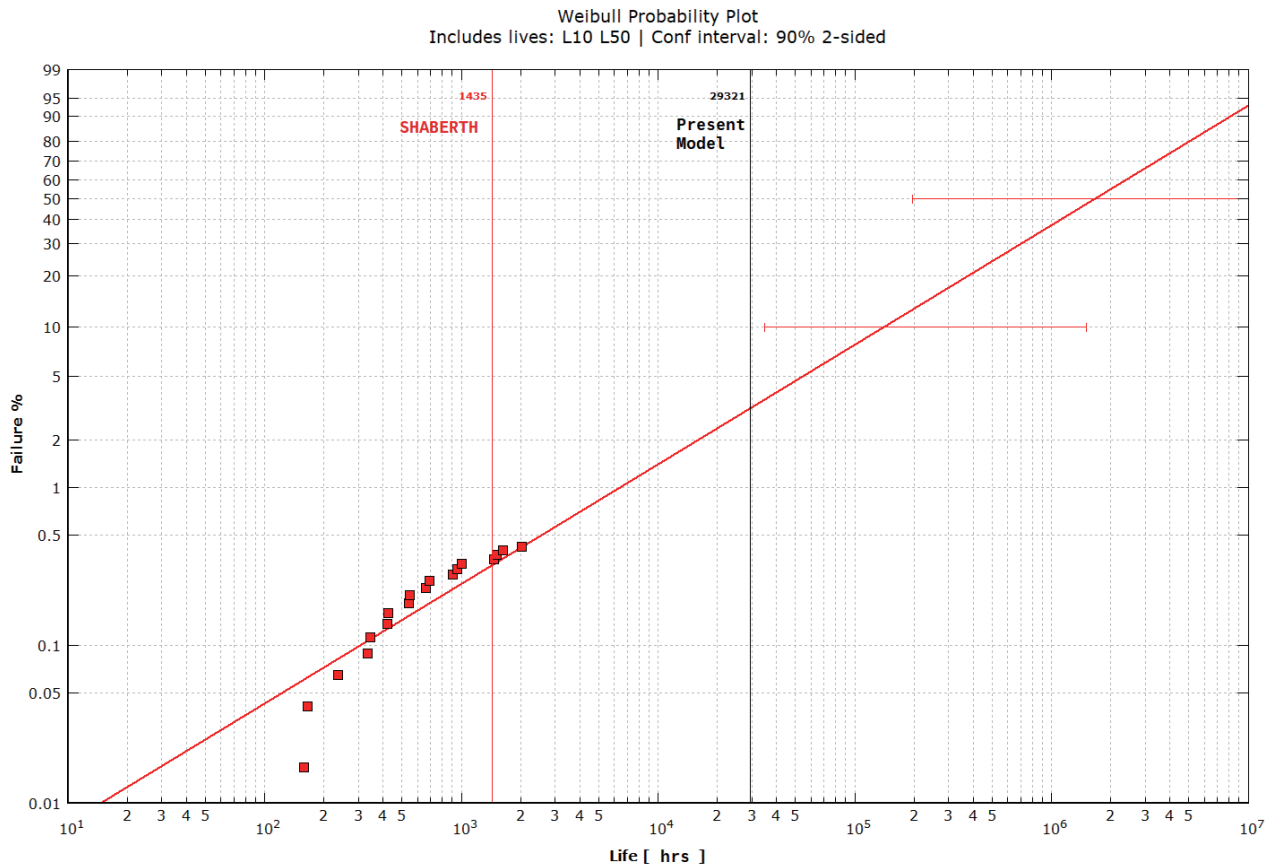


Fig. 9 Weibull plot corresponding to the failure incidents presented in [1] with a sample of 4173 bearings compared with the calculated life with the here presented model (black vertical line). For reference the SHABERTH results, as given in [1] is also included (red vertical line)

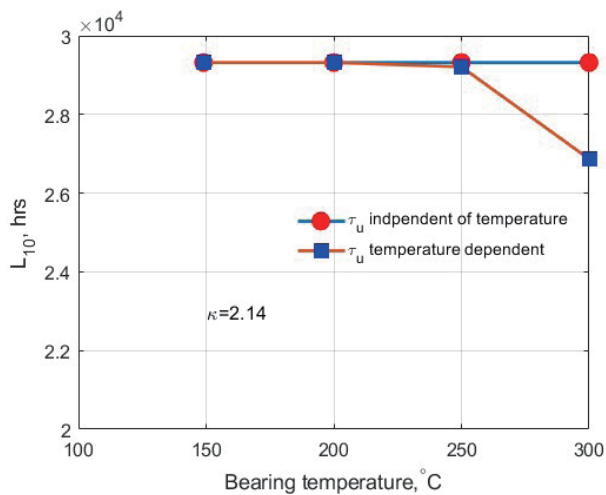


Fig. 10 Calculated life with the current model for the case described in reference [1] (AMS M50 VIM/VAR) for increasing temperature but keeping the lubrication parameter constant  $\kappa = 2.14$  for the case where the fatigue limit is reduced with increasing temperature compared with the case where this effect is disregarded

reduction of the fatigue limit of the bearing steel. Different steels have different fatigue behaviour with temperature, as demonstrated with ultrasound testing techniques in [13]. For example, AMS M50 VIM/VAR steel has higher capacity to retain its fatigue strength at high temperature as compared to ASTM 52100. This also has been demonstrated in actual endurance testing at high temperatures with the results summarized here. After validation, the model introduced here when compared with available field experimental information for AMS M50 VIM/VAR steel gives good results, it predicts higher bearing life than the currently used model from SHABERTH but lower life than the stress-based method introduced previously by Ioannides and Harris [2] model. When the same case is brought to even higher temperatures ( $> 250^\circ\text{C}$ ) the penalization of the fatigue limit due to high temperatures shows a reduction of the bearing life even for this material, as expected. Even when Eq. (7) at  $200^\circ\text{C}$  predicts a reduction of fatigue limit of about 15%, the reduction of life is still very small at this temperature.

In relation to the ratio of surface and subsurface damage, the parameter RSF (described in Eqs. (6)) and compared with the actual surface and subsurface failures obtained from the endurance tests. Notice that oil cleanliness in this test is normally very good, having online filtration. However, particle overrolling cannot be avoided and  $\eta_c$  is estimated to be around 0.8. Table 8 shows the summary of the comparison of the percentage of real surface failures found in the different endurance tests with the calculated RSF value from Eqs. (6). A one-to-one comparison is very difficult because the true value of

Table 8 Failure analysis results from endurance testing compared with RSF values, as calculated from Eqs. (6)

| Test            | Outer ring temperature, [°C] | % of surface failures | % of surface failures non-indentation related | Calculated RSF in % |
|-----------------|------------------------------|-----------------------|-----------------------------------------------|---------------------|
| AMS 52100 VAR   | 90, "low"                    | 85                    | 55                                            | 50                  |
| AMS 52100 VAR   | 200, "high"                  | 7                     | 0                                             | 50                  |
| AMS M50 VIM/VAR | 90, "low"                    | 37                    | 6                                             | 50                  |
| AMS M50 VIM/VAR | 200, "high"                  | 100                   | 20                                            | 50                  |

$\eta_c$  is likely to change from test to test. However, Table 8 shows the percentage of the total surface originated failures and as well the percentage of surface originated failures unrelated to indentations from particle overrolling. The true RSF values will be represented somewhere in between these two percentages. Depending on which column is considered, in some cases the calculated value seems to disagree with the actual results from the test and in some cases the agreement is reasonable. The result is not surprising in endurance testing, where small variation in cleanliness conditions can have large consequences in bearing failure distribution.

In general, it can be said that the advantage of using this model, apart from the consideration of new steels with higher fatigue strength in high temperature conditions is that, since it separates the surface and the subsurface survival it is a highly suitable model to model the life of ceramic (hybrid) bearings [8, 9] and the effect of sporadic indentations [10], often present in these applications. When combinations of high load and high speed are present in rolling bearings, the identification of the boundary of the fatigue failure and the seizure failures modes [11] (called transition diagram) becomes very relevant in the application of the relevant model for the proper expected failure mode. In the current case, a conservative line has been obtained via testing for both executions, hybrid and steel-steel bearings. The failure mode of adhesive wear (seizure) has not been included in the present model, however investigations in this direction have been pursued elsewhere [17].

## 7 Conclusions

The effect of high temperatures on fatigue of bearing steels has been included, to penalize the bearing life with the use of a previously developed bearing life model that separates surface from subsurface survival. Bearing endurance tests were carried out at high temperature with the reference steel AMS 52100 VAR and the high-temperature aero steel AMS M50 VIM/VAR, to validate the model. Anti-seizure minimum life tests were conducted to identify a conservative boundary of operating conditions, where the fatigue model is applicable. The model was compared with published field mainshaft engine bearing life results and with the aero-industry bearing life model of SHABERTH. The present model gives better agreement with the field results.

From this analysis, the following conclusions can be drawn:

1. The presented model gives good results against high-temperature endurance tests and previously published field results.
2. The presented model identifies a boundary of application conditions based on testing to impose applicability limits of it.
3. The fact that this model separates surface from subsurface survival gives the possibility to properly model hybrid

bearings and initial sporadic indentation damage often present in these applications.

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## Nomenclature

| Symbol         | Description                                                     | Units          |
|----------------|-----------------------------------------------------------------|----------------|
| $a_2$          | Material factor, SHABERTH model                                 | -              |
| $a_3$          | Lubrication factor, SHABERTH model                              | -              |
| A              | Area                                                            | m <sup>2</sup> |
| $\bar{A}$      | Constant                                                        | -              |
| $\bar{B}$      | Constant                                                        | -              |
| c              | Exponent                                                        | -              |
| e              | Weibull slope                                                   | -              |
| h              | Exponent                                                        | -              |
| $h_{\min}$     | Lubricant minimum film thickness                                | m              |
| $L_{10}$       | Bearing life with 90% reliability                               | hrs            |
| $L_{10.50}$    | 50 % percentile of $L_{10}$                                     | hrs            |
| $R_a$          | Composed arithmetic average roughness                           | m              |
| $R_q$          | Composed r.m.s. of the roughness (two surfaces)                 | m              |
| u              | Number of overrollings per revolution                           | -              |
| S              | Reliability                                                     | -              |
| <sub>sur</sub> | Relates to surface                                              | -              |
| <sub>sub</sub> | Relates to subsurface                                           | -              |
| T              | Temperature                                                     | °C             |
| $V_v$          | Stressed volume                                                 | m <sup>3</sup> |
| z              | Depth coordinate                                                | m              |
| $\kappa$       | Viscosity ratio, as defined in ISO 281:2007                     | -              |
| $\Lambda$      | Relative film thickness, $\Lambda = h_{\min}/R_q$               | -              |
| $\sigma_v$     | Stress amplitude used in the fatigue criterion                  | Pa             |
| $\sigma_{v,u}$ | Fatigue limit corresponding to $\sigma_v$                       | Pa             |
| $\tau_v$       | Orthogonal shear stress amplitude used in the fatigue criterion | Pa             |
| $\tau_{v,u}$   | Fatigue limit corresponding to $\tau_v$                         | Pa             |



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